Financial Network & Energy Crisis in the S&P 500

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Introduction

Escalated conflicts in Iran led to increasing oil prices and affected many sectors. This poses the question of how this conflict affects financial market structures.

This project aims to understand the overall network dynamics and movement of the S&P 500 stocks. The main objective of this project is to explore new machine learning methods and how they can be applied to provide meaningful insights in financial settings. Data is downloaded from Yahoo Finance for quick access to daily stock prices. By using only correlations of returns, the analysis provides a quick way to access the co-movement of stocks. When combined with network analysis techniques, it is easy to see how the overall network behaves following changes in international politics.

The results confirms that stocks are less distinctive in crisis times, as sectors are clustered closely. Of all three periods, the energy sector is the most separated from other sectors. However, since the analysis is based on a very short period, it does not establish causal relationships, but instead just offers insights into co-movements and evaluates the use of machine learning.

Data & Methods

I chose the start of the Iran conflict (Feb 28, 2026) as a 'crisis' and separated my time periods accordingly. Timeframe:

- 1-year pre-crisis: Feb 27, 2025, to Feb 27, 2026
- Pre-crisis: Jan 1, 2026, to Feb 27, 2026
- Post-crisis: Feb 28, 2026, to April 24, 2026

Data sources: yfinance & list of tickers/sectors from https://datahub.io/core/s-and-p-500-companies

Data Processing

- Downloaded closing prices from yfinance (from Feb 27, 2025, to Apr 24, 2026)
- Calculated 1-year returns to choose the top 5 stocks from each sector – ends up with a set of 55 stocks.
- Those stocks are then monitored to see how they are changing after the conflict.

Network Analysis

- Pairwise correlations between stocks (returns) are used to construct the network.
- NetworkX is used to help with visualization.
- Any two stocks with a correlation > 0.5 are defined as 'connected,' and an edge is drawn between these two nodes.
- The size of each node is shown as the number of connections it has.

Community Detection (unsupervised ML)

Using the Louvain method, communities are detected for the 1-month pre-conflict and post-conflict comparison. This helps see how stocks are clustered together in the most optimal way.

Community Detection

Based on the given network graphs, the Louvain method aims to maximize modularity (make node clusters most distinctive from each other). It's a 2-step process:

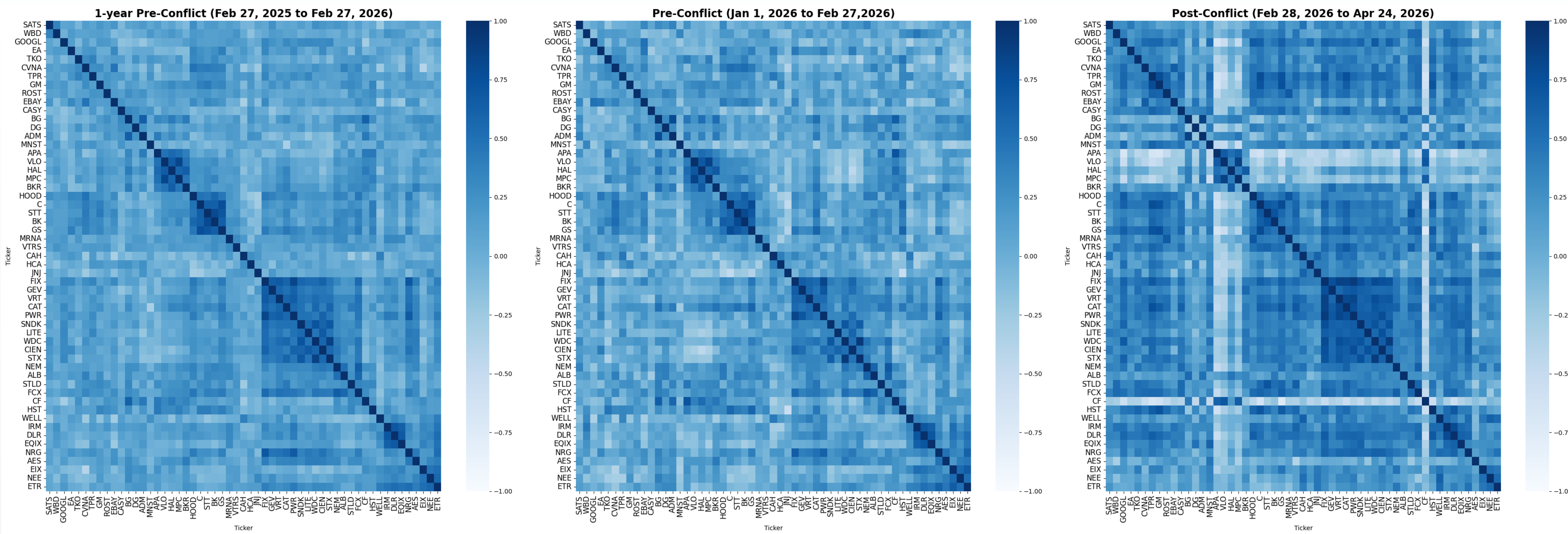
- Local optimization phase: each node is treated as its own community.
- Aggregation phase: group into "super-nodes."

The total number of communities fell from 25 to 10 after the conflict.
Community 5 has stocks from Healthcare, Tech, Real Estate, etc., all together.
Community 10 has even more sectors in it.

Post-conflict Communities

Community	Stocks
1	AES
2	APA, CF, VLO, BKR, HAL, MPC
3	BG, ALB, ADM
4	DG
5	CVNA, GOOGL, HCA, EBAY, DLR, IRM, HOOD, MRNA
6	SATS
7	EA
8	NEE, WELL, JNJ, CAH, EIX, ETR
9	PWR, STX, NRG, CIEN, WDC, FIX, SNDK, VRT, GEV, EQIX, LITE
10	NEM, GS, ROST, WBD, GM, C, MNST, STT, TPR, TKO, HST, BK, VTRS, STLD, CASY, CAT, FCX

Analysis - Correlations

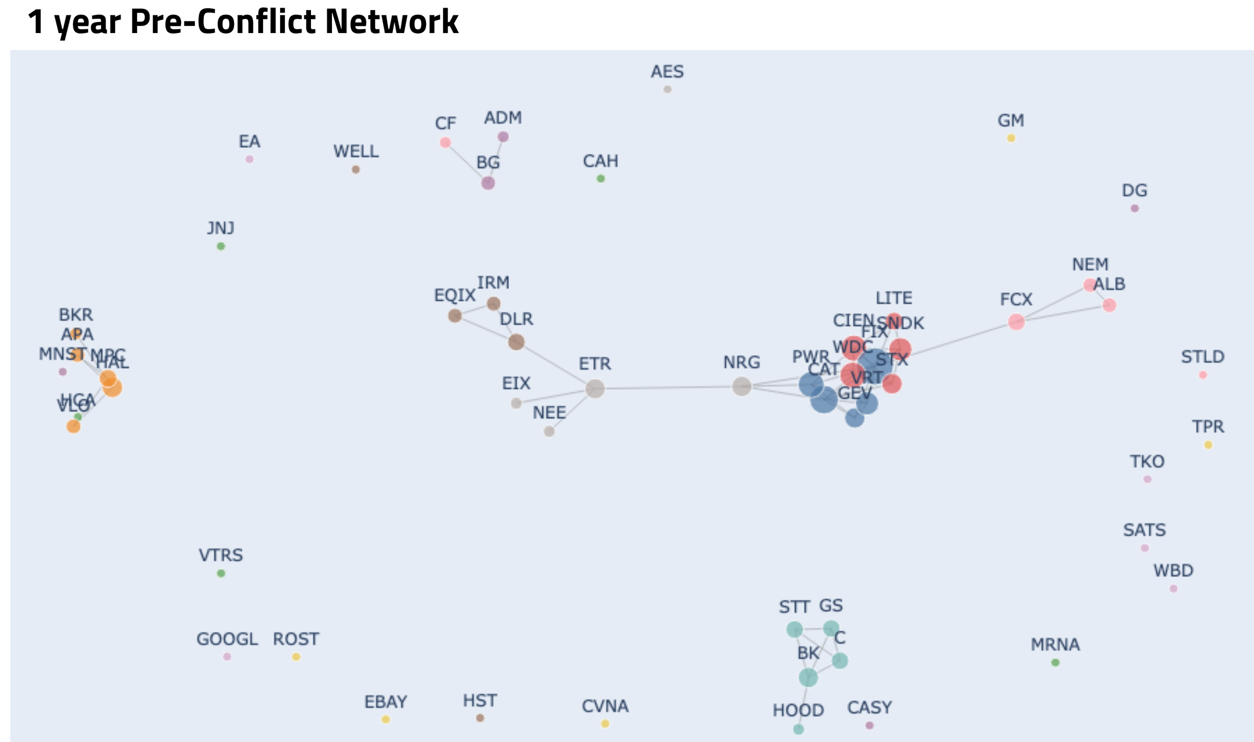


Correlations of the 1-year mark and the 1-month pre-crisis periods are very similar, while post-conflict correlation is much denser. This shows that after the conflict, stocks are moving more similarly and less distinctly.

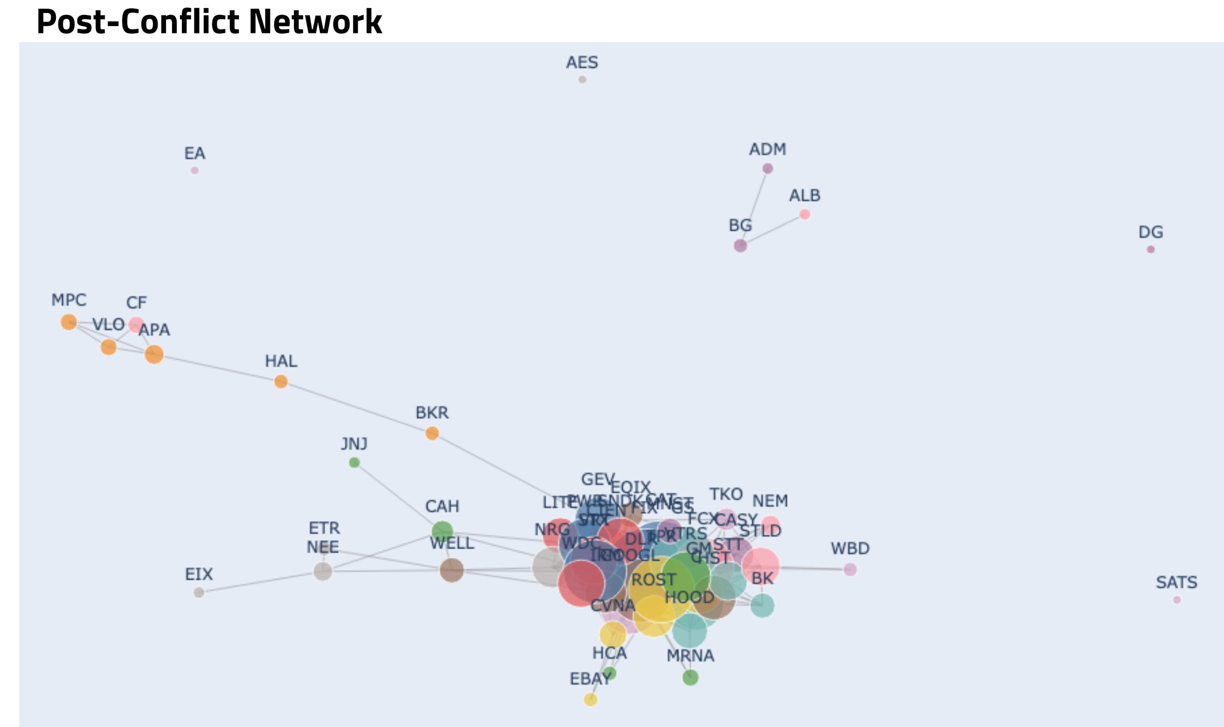
Shift from idiosyncratic risk to market-wide risk.

Results

1 year Pre-Conflict Network



Post-Conflict Network



1 month Pre-Conflict Network

By comparing the 1-year mark and the 1-month mark before the conflict, we can see the fundamental structure of all the chosen stocks. Stocks separate well by their sector in periods before the conflict.

After the conflict, network clusters together, while energy remains in its own cluster. The size of each node (which indicates the number of connections each stock has) also grows.

- Consumer Discretionary and Real Estate join market-wide movement instead of going separately as they did before.
- Resistant stocks: Consumer Staples (SATS, DG), Energy sectors, Utilities (EIX)

Conclusion

The Iran conflict triggered a structural shift in S&P 500 co-movement, captured through 55 representative stocks. Post-conflict, cross-sector correlations rose sharply, collapsing previously distinct communities into a single dominant cluster.

Energy stands as an exception for this disruption. As a direct beneficiary of conflict-driven oil price changes, energy stocks are most likely to not converge with the broader market and tend to move in the opposite direction.

As a result, sector diversification is no longer effective for managing risk during this period.

Limitation

- Periods might be too short to draw definitive conclusion on stock connection in the long run.